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(54) **INTERNAL COMMUNICATIONS WITHIN A NETWORK DEVICE**

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H04L 45/74 (2022.01)
H04L 47/62 (2022.01)

(52) **U.S. Cl.**
CPC **H04L 69/22** (2013.01); **H04L 45/74** (2013.01); **H04L 47/62** (2013.01)

(58) **Field of Classification Search**
CPC H04L 69/22; H04L 45/74; H04L 47/62
See application file for complete search history.

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(57) **ABSTRACT**

A network device forwards packets among a plurality of ports based on forwarding decisions made by a packet processor. Forwarding the packets among the plurality of ports comprises storing at least payloads of the packets in a packet memory until the packets are transmitted by the network device. Additionally, the network device communicates non-packet data within the network device, including accessing the packet memory to write the non-packet data to the packet memory, and accessing the packet memory to read the non-packet data from the packet memory. The non-packet data includes information not included in any packets received or transmitted by the network device, and for which the packet processor makes a forwarding decision.

20 Claims, 5 Drawing Sheets

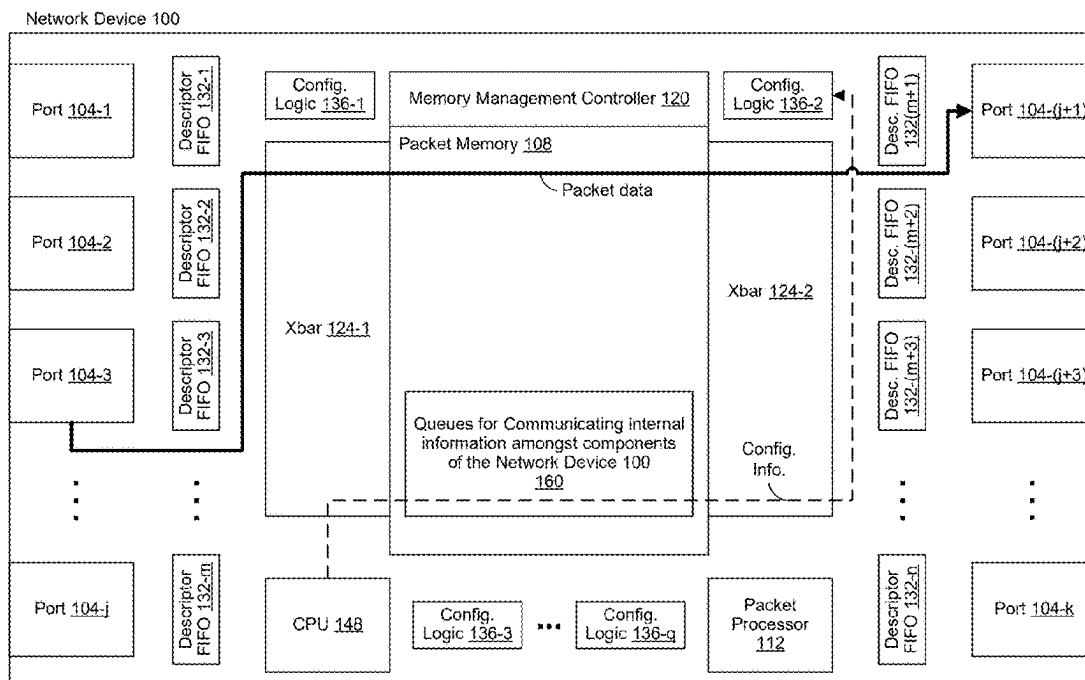


FIG. 1

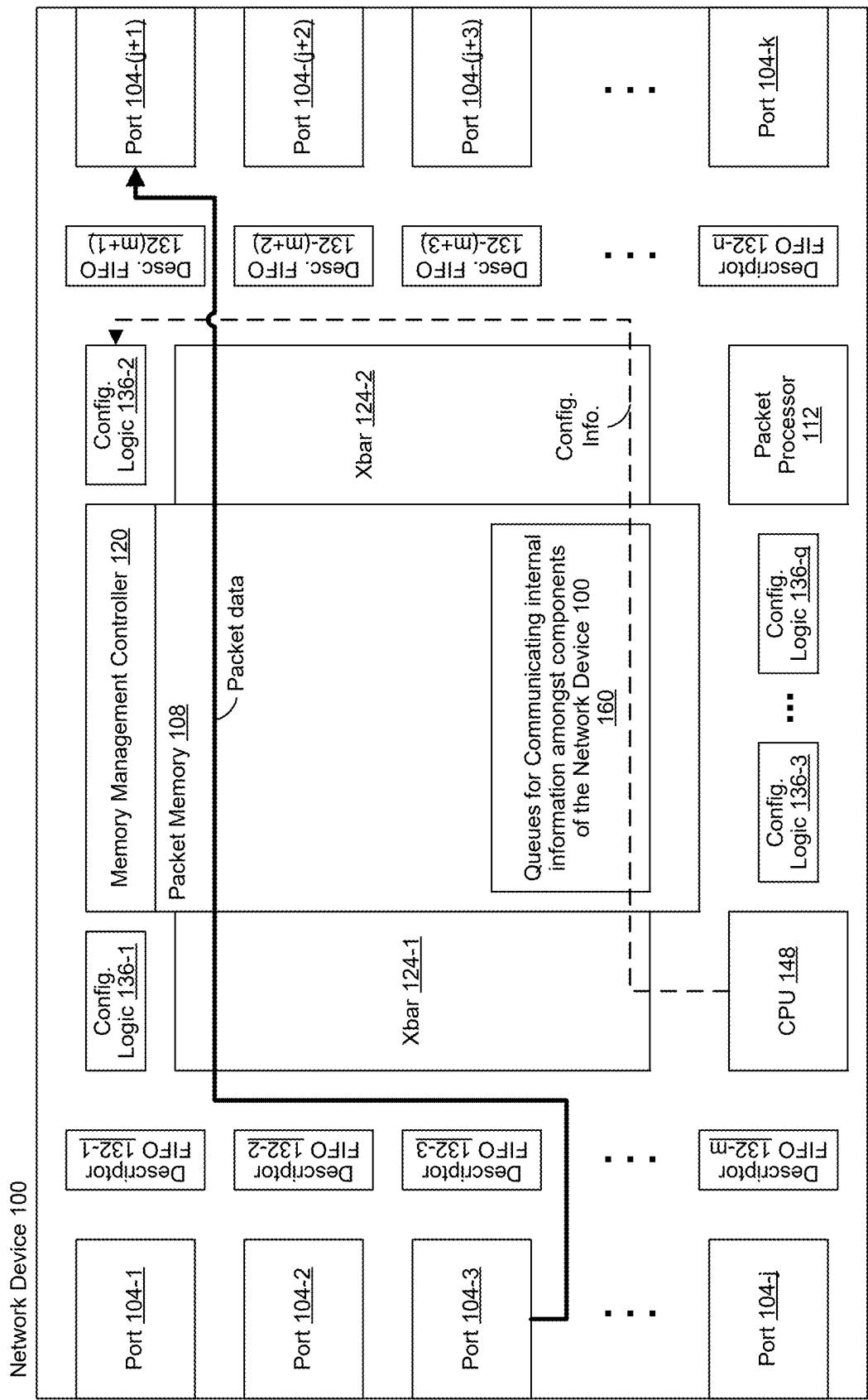


FIG. 2

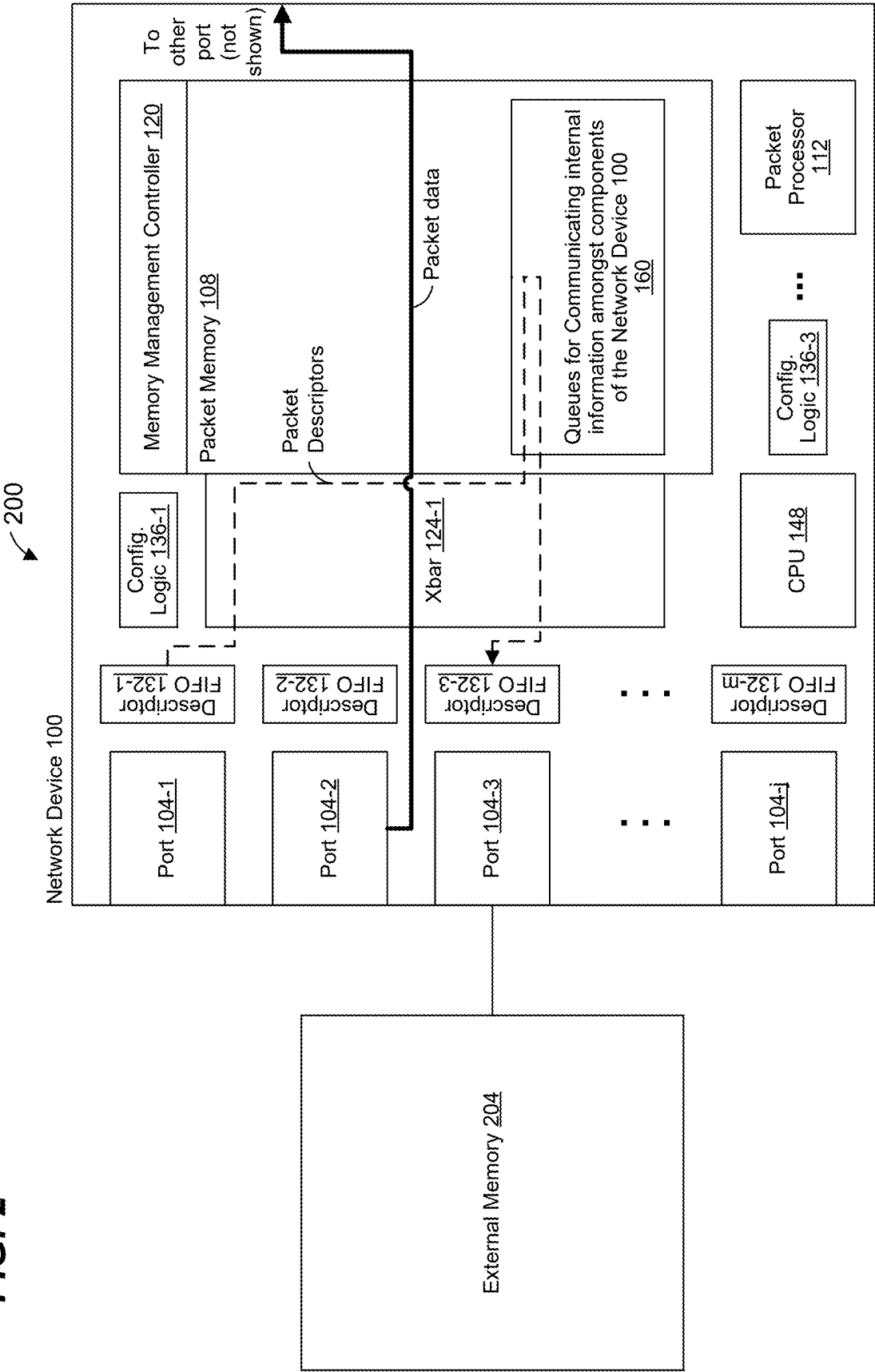


FIG. 3

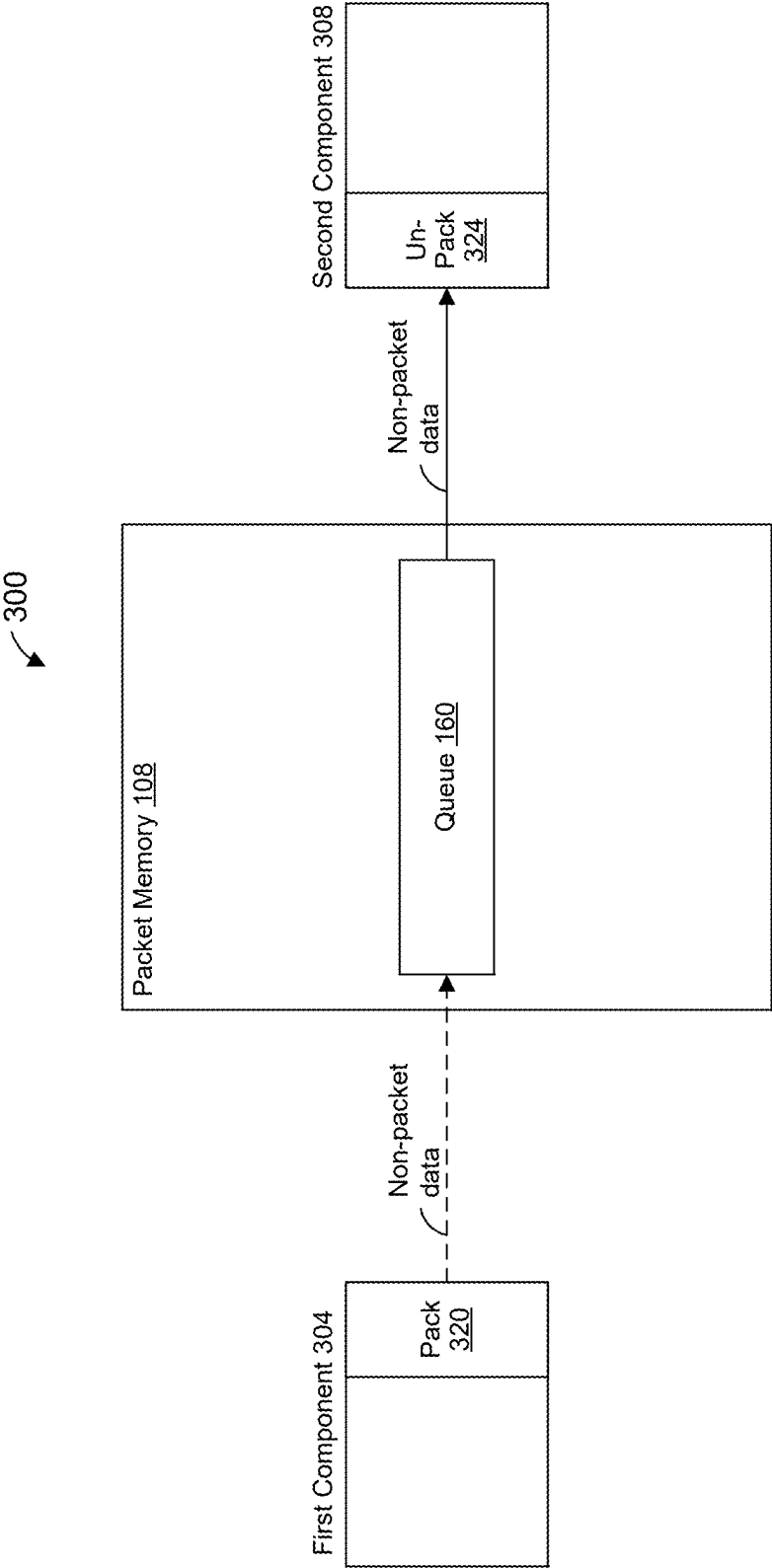


FIG. 4

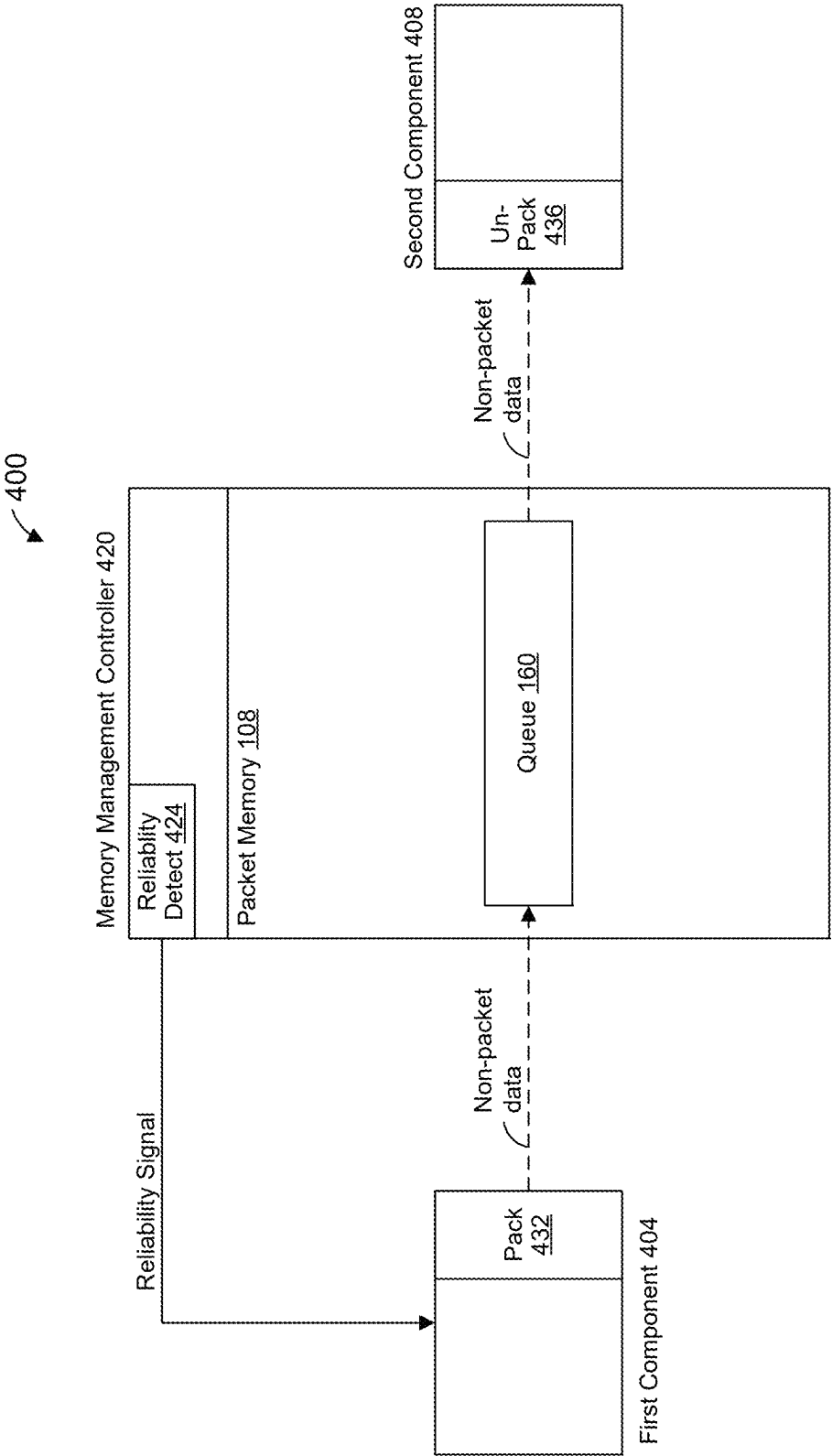
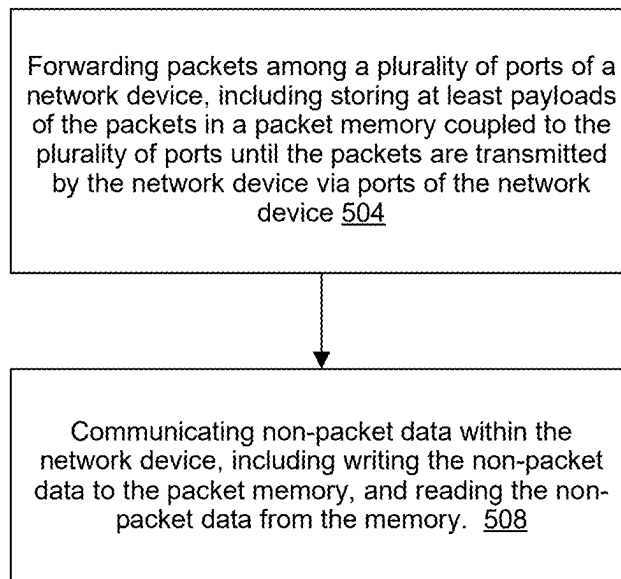


FIG. 5

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INTERNAL COMMUNICATIONS WITHIN A NETWORK DEVICE

CROSS REFERENCES TO RELATED APPLICATIONS

This application claims the benefit of U.S. Provisional Patent Application No. 63/141,452, entitled "Packet Switching for Internal Messaging," filed on Jan. 25, 2021, which is incorporated herein by reference in its entirety for all purposes.

FIELD OF TECHNOLOGY

The present disclosure relates generally to network communications, and more particularly to conveying information within a network device.

BACKGROUND

Network devices such as switches, routers, etc., typically include a packet buffer to store received packets while the network device determines ports of the network device via which the packets are to be transmitted and performs other packet processing operations on received packets. Network devices also include other components such as a central processing unit (CPU), a packet processor, queues for storing packet descriptors, etc., and such other components typically need to communicate information with other components within the network device. Thus, typical network devices include internal communication interconnection networks (e.g., busses, crossbars, etc.) for routing internally used information amongst components within the network device. Such internal interconnection networks within network devices typically consume a significant amount of integrated circuit (IC) chip area and thus are responsible for a substantial portion of the cost of IC chips of network devices. Additionally, such internal interconnection networks are responsible for a substantial portion of the power consumed by IC chips of network devices.

SUMMARY

In an embodiment, a network device comprises: a plurality of ports configured to communicatively couple to a plurality of communication links; a packet processor configured to process at least headers of packets received via ports of the network device to make forwarding decisions regarding the packets; a packet memory coupled to the plurality of ports, the packet memory configured to store at least payloads of packets received via ports of the network device until the packets are transmitted by the network device via ports of the network device; a plurality of crossbars coupled to the packet memory, each crossbar configured to provide to a respective set of ports of the network device read access and write access to the packet memory; and a plurality of non-port components associated with processing packets received by the network device, the plurality of non-port components coupled to the plurality of crossbars, each non-port component being configured to send non-packet data to one or more of i) at least one port and ii) at least one other non-port component, by storing non-packet data in the packet memory for retrieval by the one or more of i) the at least one port and ii) the at least one other non-port component, the non-packet data comprising information not included in any packet i) received or trans-

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mitted by the network device and ii) for which the packet processor makes a forwarding decision.

In another embodiment, a method for communicating information within a network device includes: forwarding packets among a plurality of ports of the network device based on forwarding decisions made by a packet processor of the network device, the packets having been received via communication links coupled to the plurality of ports and subsequently transmitted by the network device via the plurality of ports, wherein forwarding the packets among the plurality of ports comprises storing at least payloads of the packets in a packet memory coupled to the plurality of ports until the packets are transmitted by the network device via ports of the network device, including i) accessing the packet memory via a plurality of crossbars coupled to the packet memory to write the packets to the packet memory, and ii) accessing the packet memory via the plurality of crossbars to read the packets from the packet memory; and communicating non-packet data within the network device, including i) accessing the packet memory via the plurality of crossbars to write the non-packet data to the packet memory, and ii) accessing the packet memory via the plurality of crossbars to read the non-packet data from the packet memory, the non-packet data comprising information not included in any packets i) received or transmitted by the network device and ii) for which the packet processor makes a forwarding decision.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a simplified diagram of an example network device in which non-packet data is communicated within the network device via a packet memory of the network device, according to an embodiment.

FIG. 2 is a simplified block diagram of an example system in which the network device of FIG. 1 uses a packet memory to communicate packet descriptors between components of the network device, according to an embodiment.

FIG. 3 is a simplified diagram of an example system within a network device for communicating non-packet information via a packet memory of the network device, according to an embodiment.

FIG. 4 is a simplified diagram of another example system within a network device for communicating non-packet information via a packet memory of the network device, according to another embodiment.

FIG. 5 is a flow diagram of an example method for communicating non-packet information within a network device using a packet memory of the network device, according to an embodiment.

DETAILED DESCRIPTION

In embodiments described below, a network device includes a packet memory for storing received packets while the network device determines ports of the network device via which the packets are to be transmitted. In some embodiments, the size and/or access rate of the packet memory is designed to handle relatively infrequent events, such as bursts of packets (i.e., a high number of packets received in a short amount of time), relatively long packets, etc. As a result, the packet memory often has excess capacity during operation, e.g., bandwidth availability when a rate at which the packet memory is being accessed to store packets to the packet memory and read packets from the packet memory is below peak levels, an amount of memory space not currently being used for storing packets, etc.

In embodiments described below, other components of the network device, such as a central processing unit (CPU), a packet processor, queues for storing packet descriptors, etc., utilize the excess capacity of the packet memory to communicate non-packet information to other components within the network device. As used herein, the term “non-packet information” refers to data that is not included any network packet i) that is received or transmitted by the network device and ii) for which the packet processor makes a forwarding decision. As used herein, the term “network packet” refers to a formatted unit of data (a “packet”) that includes one or more headers having network information that network devices use to route the packet through a communication network.

As an illustrative example, the packet memory is configured to store queues corresponding to respective pairs of components within the network device, and the respective pairs of components utilize the queues to communicate non-packet information. By using queues within the packet memory for communicating non-packet information within the network device, integrated circuit (IC) chip area devoted to internal communication interconnection networks (e.g., busses, crossbars, etc.) for communicating such non-packet information amongst components within the network device is reduced as compared to typical network devices, according to some embodiments. Because such internal interconnection networks are reduced as compared to typical network devices, power consumption is also reduced as compared to typical network devices, at least in some embodiments.

FIG. 1 is a simplified diagram of an example network device 100, according to an embodiment. The network device 100 includes a plurality of ports 104, a packet memory 108, and a packet processor 112. At least some of the ports 104 are configured to communicatively couple to respective communication links. Packets received via communication links coupled to ports 104 are temporarily stored in the packet memory 108 while the packet processor 112 processes at least headers of the packets to determine ports 104 via which the packets are to be transmitted.

Components of the network device 100, such as the ports 104 and other components to be described below, access the packet memory 108 via a memory management controller 120 and one or more crossbars 124. Although two crossbars 124 are illustrated in FIG. 1, the network device 100 includes any suitable number of crossbars 124, such as one, three, four, five, etc., in various embodiments. Generally, respective sets of components of the network device 100 access the packet memory 108 via respective crossbars 124, in some embodiments.

In some embodiments, the packet memory 108 is a part of a larger memory device that stores information used by the network device 100 in addition to the information stored in the packet memory 108. Examples of information stored in the larger memory device in addition to the information stored in the packet memory 108 are described below. In other embodiments, the packet memory 108 corresponds to a first memory device, and information used by the network device 100 in addition to the packet memory 108 is stored in one or more second memory devices that are distinct from the first memory device. In some embodiments, the packet memory 108 corresponds to multiple memory devices.

In some embodiments, the size and/or access rate of the packet memory 108 is designed to handle relatively infrequent events, such as bursts of packets (i.e., a high number of packets received in a short amount of time), receipt of relatively long packets, etc. As a result, the packet memory

108 often has excess capacity during operation, e.g., an amount of memory space not currently being used for storing packets, access availability when the packet memory 108 is not being accessed to store packets to the packet memory 108 or read packets from the packet memory 108, etc.

When a packet is received via one of the ports 104, the memory management controller 120 allocates memory resources in the packet memory 108 for storing the packet. Additionally, the memory management controller 120 generates a packet descriptor corresponding to the packet, and provides the packet descriptor to the packet processor 112 for processing by the packet processor 112. Packet descriptors and processing of packet descriptors by the packet processor 112 will be discussed in more detail below.

A received packet (or at least a data portion of the receive packet) is routed from the port 104 at which the packet was received to the packet memory 108 via a crossbar 124 that is coupled to the port 104, and the packet is stored in the packet memory 108 using memory resources allocated by the memory management controller 120.

In an embodiment, the packet memory 108 is organized as a plurality of buffers of one or more fixed lengths, and the memory management controller 120 allocates linked lists of buffers for storage of packet data. For example, in connection with a packet being received at one of the ports 104, the memory management controller 120 allocates a linked list of buffers within the packet memory 108 for storing the received packet. In other embodiments, the packet memory 108 is organized in another suitable manner and the memory management controller 120 allocates memory space in the packet memory 108 for storage of packet data in another suitable manner.

As discussed above, the memory management controller 120 generates packet descriptors corresponding to respective received packets. The packet descriptor includes header information extracted from the packet by the memory management controller 120, such as Layer-2 header information, Layer-3 header information, tunnelling header information, etc. The packet descriptor also includes information that is associated with the packet but that is not included within any headers of the packet when the packet is received by the network device 100 and is not included within any headers of the packet when the packet is transmitted by the network device 100 (referred to herein as “associated packet information”). Examples of associated packet information that may be included in the packet descriptor by the memory management controller 120 include an indication of a location in the packet memory 108 at which the packet is stored, an identifier (ID) of the port 104 via which the packet was received (“source port ID”), an indicator of a time at which the packet was received by the network device 100 (“receive time indicator”), etc. Additionally, the packet processor 112, when processing the packet descriptor, may add associated packet information to the packet descriptor such as an indicator of one or more ports 104 via which the packet is to be transmitted (“target ports”), an indicator of a priority level according to which the packet is to be transmitted by the target port(s), an indicator of whether the packet is to be dropped, an indicator of whether a tunneling header is to be added to the packet, an indicator of whether a tunneling header is to be removed from the packet, an indicator of whether a timestamp is to be added to the packet prior to transmitting the packet, etc.

In some embodiments, the packet processor 112 comprises a Layer-2 bridge engine configured to process at least Layer-2 headers (e.g., media access control (MAC) headers)

of packets received via the ports **104** to determine ports **104** via which the packets are to be transmitted. In embodiments in which the packet processor **112** comprises a Layer-2 bridge engine, the Layer-2 bridge engine uses at least Layer-2 header information in a packet descriptor to determine one or more target ports via which a packet corresponding to the packet descriptor is to be transmitted and stores indicator(s) of the determined target ports in the packet descriptor. In some embodiments in which the packet processor **112** comprises a Layer-2 bridge engine, the Layer-2 bridge engine utilizes a forwarding table that stores associations between i) at least Layer-2 information (e.g., destination MAC addresses, VLAN identifiers (VIDs), etc.) and ii) ports **104** to determine one or more target ports via which a packet corresponding to the packet descriptor is to be transmitted. In some embodiments, the forwarding table is stored in one or more memory devices that are distinct from one or more other memory devices corresponding to the packet memory **108**. In other embodiments, the forwarding table is stored in one or more memory devices corresponding to the packet memory **108**.

In some embodiments, the packet processor **112** additionally or alternatively comprises a Layer-3 routing engine configured to process at least Layer-3 headers (e.g., Internet Protocol (IP) headers) of packets received via the ports **104** to determine ports **104** via which the packets are to be transmitted. In embodiments in which the packet processor **112** comprises a Layer-3 routing engine, the Layer-3 routing engine uses IP header information in a packet descriptor to determine a next hop IP address to be included in a packet corresponding to the packet descriptor, and determines one or more target ports via which the packet is to be transmitted. Additionally, the Layer 3 routing engine stores the next hop IP address and indicator(s) of the determined target ports in the packet descriptor. In some embodiments in which the packet processor **112** comprises a Layer-3 routing engine, the Layer-3 routing engine utilizes a forwarding table that stores Layer-3 forwarding information and associations with ports **104** to determine a next hop IP address and one or more target ports via which a packet corresponding to the packet descriptor is to be transmitted. In some embodiments, the forwarding table is stored in one or more memory devices that are distinct from one or more other memory devices corresponding to the packet memory **108**. In other embodiments, the forwarding table is stored in one or more memory devices corresponding to the packet memory **108**.

When the packet processor **112** completes processing of a packet descriptor, the packet descriptor is stored in one or more first-in, first-out (FIFO) buffers **132** (descriptor FIFOs **132**) corresponding to one or target ports **104** that the packet processor **112** determined and via which the packet is to be transmitted. In connection with a packet descriptor reaching a head of a descriptor FIFO **132**, the port **104** that corresponds to the descriptor FIFO **132** uses information stored in the packet descriptor (e.g., an indication of a location in the packet memory **108** at which a packet corresponding to the packet descriptor is stored) to retrieve the packet, and transmits the packet via the communication link coupled to the port **104**. Each port **104** retrieves packets from the packet memory **108** via a crossbar **124** that is coupled to the port **104**.

FIG. 1 illustrates, as an example, packet data (e.g., at least a payload of a packet) that was received at port **104-3** being stored in the packet memory **108**, and the packet data subsequently being read out from the packet memory **108** and provided to the port **104-(j+1)** for transmission.

When a packet is retrieved from the packet memory **108** for transmission, the memory management controller **120** subsequently designates the memory resources that had been used for storing the packet as “unused”, i.e., the memory management controller **120** returns the memory resources to a pool of unused memory resources that are available for allocation for storing newly received packets or other information, for example.

In some embodiments, a FIFO buffer **132** is expanded into an external memory (not shown) when the FIFO buffer **132** experiences congestion (e.g., a rate at which packet descriptors are stored to the FIFO buffer **132** significantly exceeds a rate at which packet descriptors are removed from the FIFO buffer **132**, causing the FIFO buffer **132** to grow in size). In some embodiments, the external memory (not shown) is coupled to one of the ports **104**, and packet descriptors in the FIFO buffer **132** that is experiencing congestion are transferred from the FIFO buffer **132** to the one port **104** for transfer to and temporary storage in the external memory. When packet descriptors in the external memory are nearing a head of the FIFO buffer **132**, the packet descriptors are retrieved from the external memory and provided to the one port **104**, and then transferred from the one port **104** back to the FIFO buffer **132**.

The network device **100** also includes configuration logic blocks **136** that control operation of various components of the network device **100**. As an illustrative example, respective configuration logic blocks **136** correspond to each of at least some of the ports **104**, each such configuration logic block **136** controlling operation of the respective port **104**, according to an embodiment. For example, the configuration logic block **136** for a respective port **104** controls one or more of (or none of): i) whether the port **104** is to timestamp packets being transmitted by the port, ii) whether the port **104** is to timestamp packets received by the port, iii) whether the port **104** is to count bytes being transmitted by the port, iv) whether the port **104** is to count bytes received by the port, v) a transmit/receive rate at which the port is to operate, vi) a protocol the port is to use, vi) an error correction code the port is to use, etc. As another illustrative example, one or more configuration logic blocks **136** correspond to the packet processor **112** and control operation of the packet processor **112**.

Each of at least some of the configuration logic blocks **136** includes respective memory (not shown) that stores configuration information used by the configuration logic block **136** to control operation of a respective component of the network device **100**.

The network device **100** also includes a central processing unit (CPU) **148** that manages operation of the network device **100**, according to an embodiment. The CPU **148** includes, or is coupled to, a memory (not shown) that stores machine readable instructions (e.g., software, firmware, etc.) that, when executed by the CPU **148**, causes the CPU **148** to perform various acts associated with management of operation of the network device **100**. For example, in some embodiments, the CPU **148** controls operation of various components of the network device **100** by generating respective configuration information and sending the respective configuration information to respective configuration logic blocks **136**. In an embodiment, configuration information generated by the CPU **148** is stored in memories corresponding to the configuration logic blocks **136**.

In some embodiments, the CPU **148** sends one or more of management information, status information, debugging

information, etc., to an external CPU (not shown) coupled to one of the ports **104** or to a dedicated interface distinct from the ports **104**.

In some embodiments, various components of the network device **100** send information to the CPU **148** that the CPU **148** uses to control and/or manage operation of the network device **100**. Examples of information sent by components of the network device **100** to the CPU **148** include status information, management information, debugging information, etc., associated with operation of the components of the network device **100**.

As discussed above, non-packet information that is internally used by the network device **100** (e.g., configuration data, status information, debugging information, associated packet information within packet descriptors (e.g., receive port IDs, indicators of target ports, locations of packets within the packet memory **108**, etc.), etc.) are communicated amongst components of the network device **100**.

As discussed above, typical network devices include internal communication interconnection networks (e.g., busses, crossbars, etc.) for routing such non-packet information amongst components within the network device. Such internal interconnection networks within network devices typically consume a substantial amount of integrated circuit (IC) chip area and/or are responsible for a substantial portion of the power consumed by IC chips of network devices.

As will be described further below, the network device **100** utilizes the packet memory **108** to communicate non-packet information among components of the network device **100**. By using the packet memory **108**, IC chip area dedicated to internal communication interconnection networks (e.g., busses, crossbars, etc.) for communicating such non-packet information amongst components of the network device **100** is reduced as compared to typical network devices, according to some embodiments.

The packet memory **108** stores queues **160** that are used for communicating non-packet information among components of the network device **100**. In an embodiment, the CPU **148** and/or the memory management controller **120** (and/or another suitable component of the network device **100**) configures the queues **160** for use by components of the network device **100** and informs the components of the network device **100** of the queues **160** that are to be used. For example, the CPU **148** and/or the memory management controller **120** (and/or another suitable component of the network device **100**) sets up a particular queue **160** to be used by a first component to transfer non-packet information to a second component, and informs the first component and the second component of the particular queue **160** that they are to use.

In an embodiment in which the memory management controller **120** allocates linked lists of buffers, the queues **160** comprise linked lists of buffers. In such embodiments, when a queue **160**, which comprises a linked list of buffers, is set up to permit a first component to transfer non-packet information to a second component, the memory management controller **120** informs the first component and the second component of an indicator(s) of the linked list. When the first component is to transfer non-packet information to the second component, the non-packet information is written to a head of the linked list; and the memory management controller **120** allocates a new buffer for the linked list and updates the head to point to the new buffer. Similarly, the second component retrieves from the tail of the linked list non-packet information that came from the first component; and the memory management controller **120** updates the tail

to point to a next buffer in the linked list and deallocates the buffer that was previously the tail.

In some embodiments, when a linked list of buffers (a queue **160**) is set up to permit a first component to transfer non-packet information to a second component, the memory management controller **120** informs the first component of an indicator of a head of the linked list, and informs the second component of an indicator of a tail of the linked list.

FIG. **1** illustrates (with a dashed line) the CPU **148** communicating configuration information to the configuration logic **136-2** via a queue **160** within the packet memory **108**. The queue **160** being used by the CPU **148** and the configuration logic **136-2** was previously set up by the CPU **148** and/or the memory management controller **120** (and/or another suitable component of the network device **100**), and the configuration logic **136-2** was previously informed of the queue **160** being used by the CPU **148** to send configuration information to the configuration logic **136-2**. If the queue **160** was not set up by the CPU **148**, another component informs the CPU **148** of the queue **160** that the CPU **148** is to use to send configuration information to the configuration logic **136-2**.

The CPU **148** accesses the packet memory **108** via the crossbar **124-1** to write the configuration information to the queue **160**. Similarly, the configuration logic **136-2** accesses the packet memory **108** via the crossbar **124-2** to read the configuration information from the queue **160**.

FIG. **2** is a simplified block diagram of an example system **200** comprising a network device that uses a packet memory to communicate packet descriptors between components of the network device, according to an embodiment. The system **200** includes the network device **100** of FIG. **1**. Some components of the network device **100** of FIG. **1** are not shown in FIG. **2** merely to avoid obscuring relevant elements of the network device **100** discussed with reference to FIG. **2**.

FIG. **2** illustrates, as an example, packet data (e.g., at least a payload of a packet) that was received at port **104-2** being stored in the packet memory **108**, and the packet data subsequently being read out from the packet memory **108** and provided to another port (not shown) for transmission.

The system **200** includes a memory **204** that is external to the network device **100** ("external memory **204**") and coupled to port **104-3** of the network device **100**. The network device **100** is configured to write to and read from the external memory **204** via the port **104-3**. In the embodiment illustrated in FIG. **2**, the descriptor FIFO **132-1** has been expanded to store some packet descriptors in the external memory **204**. Thus, some descriptors are removed from the descriptor FIFO **132-1** and are temporarily stored in the external memory **204** by way of the port **104-3**.

In the embodiment illustrated in FIG. **2**, the descriptor FIFO **132-3** corresponds to the port **104-3**, and descriptors that are to be written to the external memory **204** via the port **104-3** are enqueued in the descriptor FIFO **132-3**. Thus, in connection with transferring descriptors from the descriptor FIFO **132-1** to the external memory **204**, the descriptors are temporarily stored in the descriptor FIFO **132-3**.

FIG. **2** illustrates (with a dashed line) the descriptor FIFO **132-1** communicating packet descriptors to the descriptor FIFO **132-3** via a queue **160** within the packet memory **108**. The queue **160** being used by the descriptor FIFO **132-1** and the descriptor FIFO **132-3** was previously set up by the CPU **148** and/or the memory management controller **120** (and/or another suitable component of the network device **100**), and the descriptor FIFO **132-1** and the descriptor FIFO **132-3**

were previously informed of the queue **160** being used by the descriptor FIFO **132-1** to send packet descriptors to the descriptor FIFO **132-3**.

The descriptor FIFO **132-1** accesses the packet memory **108** via the crossbar **124-1** to write the packet descriptors to the queue **160**. Similarly, the descriptor FIFO **132-2** accesses the packet memory **108** via the crossbar **124-1** to read the packet descriptors from the queue **160**.

Referring now to FIGS. **1** and **2**, in some embodiments in which the packet memory **108** is organized as a plurality of buffers having one or more fixed lengths, the fixed length(s) of the buffers are designed for writing and reading packet data at line speeds. In some embodiments, a fixed length of the buffers corresponds to a width of a read/write interface of the packet memory **108**, which is designed for writing and reading packet data at line speeds. Thus, in some embodiments, the fixed length(s) of buffers of the packet memory **108** is relatively long as compared to lengths of units of non-packet information that are being communicated between components of the network device **100**. Thus, in some embodiments, multiple units of non-packet information are aggregated by a first component of the network device to form a packaged set of non-packet information that is then stored within a buffer of the packet memory **108**. When a second component of the network device (for which the non-packet information is intended) retrieves the packaged set of non-packet information from the buffer of the packet memory **108**, the second component unpacks the individual units of non-packet information within the packaged set of non-packet information and uses the individual units of non-packet information.

FIG. **3** is a simplified diagram of an example system **300** within a network device for communicating non-packet information via a packet memory of the network device, according to an embodiment. In an embodiment, the system **300** is included within the network device **100** of FIG. **1** and is described with reference to FIG. **1** for explanatory purposes. In other embodiments, the system **300** is included within a suitable network device **100** different than the network device **100** of FIG. **1**. Other components of the network device are omitted from FIG. **3** merely to avoid obscuring relevant components of the network device described below. For example, the crossbar(s) **124** are not shown in FIG. **3**.

The system **300** includes a first component **304** of the network device that transfers non-packet information of the network device to a second component **308** of the network device via a queue **160** included within the packet memory **108** of the network device. In some embodiments, the first component **304** accesses the packet memory **108** via a first crossbar **124** (not shown) and the second component **308** accesses the packet memory **108** via a second crossbar **124** (not shown). In other embodiments, the first component **304** and the second component **308** both access the packet memory **108** via a same crossbar **124** (not shown).

In some embodiments, the first component **304** is the CPU **148**, and the second component **308** is another component of the network device **100**, such as a descriptor FIFO **132**, configuration logic **136**, a port **104**, an external CPU (not shown in FIG. **1**) coupled to the network device **100**, etc. In some embodiments, the second component **308** is the CPU **148**, and the first component **304** is another component of the network device **100**, such as a descriptor FIFO **132**, configuration logic **136**, a port **104**, an external CPU, etc.

In various embodiments, the non-packet information transferred between the first component **304** and the second component **308** comprises management information, control

information, configuration information, non-packet information within packet descriptors, etc.

In some embodiments, the first component **304** is a descriptor FIFO **132**, and the second component **308** is another component of the network device **100**, such as another descriptor FIFO **132**, etc.

In some embodiments, the packet memory **108** is organized as a plurality of buffers having a fixed length that is relatively long as compared to lengths of units of non-packet information that is being communicated from the first component **304** to the second component **308**.

The first component **304** includes packing logic **320** that is configured to aggregate multiple units of non-packet information to form a packaged set of non-packet information. The packaged set of non-packet information from the first component **304** is then stored within a buffer corresponding to a queue **160** within the packet memory **108**.

Subsequently, the packaged set of non-packet information within the queue **160** is retrieved and provided to the second component **308**. The second component **308** includes unpacking logic **324** that is configured to unpack the multiple individual units of non-packet information within the packaged set of non-packet information, and the second component **308** can then use the individual units of non-packet information.

Referring again to FIGS. **1** and **2**, in some embodiments communicating non-packet information via the packet memory **108** provides “best effort” communication, i.e., delivery to the intended destination is not guaranteed. For example, in some embodiments, writing non-packet information to a queue **160** includes sending a request to the memory management controller **120** for writing the non-packet information to the queue **160**. In such embodiments, if the memory management controller **120** is unable to allocate memory resources for writing the non-packet information to the queue **160** (e.g., the memory management controller **120** and/or the packet memory **108** are too busy handling writing and/or reading packet data to/from the packet memory **108**, the packet memory **108** does not have unused memory resources that are available for use in queues **160**, etc.), writing the non-packet information to the queue **160** fails. On the other hand, if the memory management controller **120** is able to allocate the memory resources for writing the non-packet information to the queue **160**, the memory resources are allocated to the queue **160** and writing the non-packet information to the queue **160** proceeds. In some embodiments, the memory management controller **120** is configured to prioritize packet data over non-packet data, and the memory management controller **120** allocates memory resources to non-packet data only when memory access bandwidth is available.

As an illustrative example, in some embodiments in which the packet memory **108** is organized as a plurality of buffers and each queue **160** comprises a linked list of buffers, writing non-packet information to a queue **160** involves the memory management controller **120** allocating a new buffer to the queue **160**. In such embodiments, if the memory management controller **120** is too busy handling the writing and/or reading packet data to/from the packet memory **108** to allocate a new buffer to the queue **160**, and/or if the packet memory **108** does not have unused memory resources that are available for use in queues **160**, writing the non-packet information to the queue **160** fails. In some embodiments, if the packet memory **108** is too busy with writing and/or reading packet data to/from the packet memory **108**, writing the non-packet information to the queue **160** fails. On the other hand, if the memory management controller **120** is

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able to allocate a new buffer to the queue **160** and the packet memory **108** has bandwidth to handle the writing of non-packet information, the buffer is allocated to the queue **160** and writing the non-packet information to the queue **160** proceeds.

In some embodiments in which communicating information via the packet memory **108** provides “best effort” communication and is not reliable, non-packet information that requires reliable communication is communicated via a reliable interconnection network distinct from the packet memory **108**. In other embodiments, a network device includes control logic that enables reliable communication via the packet memory **108**.

FIG. **4** is a simplified diagram of another example system **400** within a network device for communicating non-packet information via a packet memory of the network device, according to another embodiment. In an embodiment, the system **400** is included within the network device **100** of FIG. **1** and is described with reference to FIG. **1** for explanatory purposes. In other embodiments, the system **400** is included within a suitable network device **100** different than the network device **100** of FIG. **1**. Other components of the network device are omitted from FIG. **4** merely to avoid obscuring relevant components of the network device described below. For example, the crossbar(s) **124** are not shown in FIG. **4**.

The system **400** includes a first component **404** of the network device that transfers non-packet information of the network device to a second component **408** of the network device via a queue **160** included within the packet memory **108** of the network device. In some embodiments, the first component **404** accesses the packet memory **108** via a first crossbar **124** (not shown) and the second component **408** accesses the packet memory **108** via a second crossbar **124** (not shown). In other embodiments, the first component **404** and the second component **408** both access the packet memory **108** via a same crossbar **124** (not shown).

In some embodiments, the first component **404** is the CPU **148**, and the second component **408** is another component of the network device **100**, such as a descriptor FIFO **132**, configuration logic **136**, a port **104**, an external CPU coupled to the network device **100**, etc. In some embodiments, the second component **408** is the CPU **148**, and the first component **404** is another component of the network device **100**, such as a descriptor FIFO **132**, configuration logic **136**, a port **104**, an external CPU coupled to the network device **100**, etc.

In various embodiments, the non-packet information transferred between the first component **404** and the second component **408** comprises management information, control information, configuration information, non-packet information within packet descriptors, etc.

In some embodiments, the first component **404** is a descriptor FIFO **132**, and the second component **408** is another component of the network device **100**, such as another descriptor FIFO **132**, etc.

In some embodiments, the packet memory **108** is organized as a plurality of buffers. A memory management controller **420** (similar to the memory management controller **120** of FIG. **1**) includes reliability detection logic **424** that is configured to determine times at which the packet memory **108** is capable of providing reliable communication via the queues **160** and to generate a reliability signal that indicates when the packet memory **108** is capable of providing reliable communication via the queues **160**. The first component **404** receives the reliability signal and uses the reliability signal to determine times at which the packet

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memory **108** is capable of providing reliable communication via the queues **160**. When the first component **404** has non-packet information that is to be communicated to the second component **408** via the queue **160**, and when the packet memory **108** is capable of providing reliable communication via the queue **160**, the first component **404** stores the non-packet information in the queue **160**. Subsequently, the non-packet information within the queue **160** is retrieved and provided to the second component **408**.

On the other hand, when the first component **404** has non-packet information that is to be communicated to the second component **408** via the queue **160**, but the packet memory **108** is not capable of providing reliable communication via the queue **160**, the first component **404** does not store the non-packet information in the queue **160** but rather waits for a time when the packet memory **108** is capable of providing reliable communication.

In some embodiments, the buffers have a fixed length that is relatively long as compared to lengths of units of non-packet information that is being communicated from the first component **404** to the second component **408**. In some embodiments, the first component **404** includes packing logic **432** that is configured to aggregate multiple units of non-packet information to form a packaged set of non-packet information.

In some embodiments, when the first component **404** has non-packet information that is to be communicated to the second component **408** via the queue **160**, but the packet memory **108** is not capable of providing reliable communication via the queue **160**, the first component **404** does not store the non-packet information in the queue **160** but rather the packing logic **432** aggregates the non-packet information with other non-packet information within a packaged set of non-packet information. When the packet memory **108** later becomes capable of providing reliable communication, the first component **404** stores the packaged set of non-packet information in the queue **160**.

Subsequently, the packaged set of non-packet information within the queue **160** is retrieved and provided to the second component **408**. The second component **408** includes unpacking logic **436** that is configured to unpack the multiple individual units of non-packet information within the packaged set of non-packet information, and the second component **408** can then use the individual units of non-packet information.

In some embodiments, the packing logic **432** and/or the unpacking logic **436** are omitted.

FIG. **5** is a flow diagram of an example method **500** for communicating non-packet information within a network device, according to an embodiment. In various embodiments, the example network device **100** (FIG. **1**), the example system **200** of FIG. **2**, the example system **300** of FIG. **3**, or the example system **400** of FIG. **4**, implements the method **500**. In other embodiments, the method **400** is implemented by another suitable network device.

At block **504**, a network device (e.g., the network device **100** or another suitable network device) forwards packets among a plurality of ports of the network device, the packets having been received via communication links coupled to the plurality of ports and subsequently transmitted by the network device via the plurality of ports. Forwarding packets among the plurality of ports at block **504** comprises storing at least payloads of the packets in a packet memory (e.g., the packet memory **108**) coupled to the plurality of ports until the packets are transmitted by the network device via ports of the network device. Storing at least payloads of the packets in the packet memory comprises i) accessing the

packet memory via a plurality of crossbars to write the packets to the memory, and ii) accessing the memory via the plurality of crossbars to read the packets from the memory.

At block 508, the network device (e.g., the network device 100 or another suitable network device) communicates non-packet data within the network device. Communicating the non-packet data at block 508 includes i) accessing the packet memory via the plurality of crossbars to write the non-packet data to the packet memory, and ii) accessing the memory via the plurality of crossbars to read the non-packet data from the memory. The non-packet data communicated at block 508 comprises information not included in any header of any packet received by the network device and not included in any header of any packet transmitted by the network device.

In various embodiments, communicating non-packet data at block 508 comprises communicating one of management information, control information, configuration information, non-packet information within packet descriptors, etc.

In various embodiments, communicating non-packet data at block 508 comprises one of i) a first CPU of the network device, ii) configuration logic, iii) a first descriptor FIFO, iv) a port, etc., a communicating non-packet data to one of i) a second CPU of the network device, ii) configuration logic, iii) a second descriptor FIFO, iv) a port, etc.

In an embodiment, the method 500 further comprises: allocating, by a memory management controller, memory resources of the packet memory for storing packet data; and allocating, by the memory management controller, memory resources of the packet memory for storing the non-packet data.

In another embodiment, the method 500 further comprises: for respective pairs of non-port components of the network device, managing, by the memory management controller, respective linked-lists of buffers in the packet memory that are configured to store respective non-packet data being communicated between the respective pairs of non-port components of the network device.

In another embodiment, the method 500 further comprises: generating, by logic circuitry, a signal that indicates when the packet memory is capable of providing reliable communication of non-packet data; when the signal indicates that the packet memory is capable of providing reliable communication of non-packet data, storing, by a first non-port component of the network device, non-packet data to the packet memory; and when the signal indicates that the packet memory is not capable of providing reliable communication of non-packet data, refraining, by the first non-port component, from storing the non-packet data to the packet memory.

In another embodiment, the method 500 further comprises: aggregating, by first logic circuitry of a first non-port component of the network device, multiple units of non-packet information into an aggregated set of non-packet information; storing the aggregated set of non-packet information in the packet memory; and unpacking, by second logic circuitry of a second non-port component of the network device, multiple units of non-packet information from the aggregated set of non-packet information that was retrieved from the packet memory.

Embodiment 1: A network device, comprising: a plurality of ports configured to communicatively couple to a plurality of communication links; a packet processor configured to process at least headers of packets received via ports of the network device to make forwarding decisions regarding the packets; a packet memory coupled to the plurality of ports, the packet memory configured to store at least payloads of

packets received via ports of the network device until the packets are transmitted by the network device via ports of the network device; a plurality of crossbars coupled to the packet memory, each crossbar configured to provide to a respective set of ports of the network device read access and write access to the packet memory; and a plurality of non-port components associated with processing packets received by the network device, the plurality of non-port components coupled to the plurality of crossbars, each non-port component being configured to send non-packet data to one or more of i) at least one port and ii) at least one other non-port component, by storing non-packet data in the packet memory for retrieval by the one or more of i) the at least one port and ii) the at least one other non-port component, the non-packet data comprising information not included in any packet i) received or transmitted by the network device and ii) for which the packet processor makes a forwarding decision.

Embodiment 2: The network device of embodiment 1, wherein the plurality of non-port components comprises: a processor configured to send configuration information to one or more of i) at least one port and ii) at least one other non-port component by accessing the packet memory via one of the crossbars to store the configuration information in the packet memory for retrieval by the one or more of i) the at least one port and ii) the at least one other non-port component, the configuration information including information that i) controls operation of the one or more of a) the at least one port and b) the at least one other non-port component and ii) is not transmitted by any communication links communicatively coupled to any ports of the network device; wherein the one or more of i) the at least one port and ii) the at least one other non-port component are configured to: i) access the packet memory via at least one of the crossbars to retrieve the configuration information from the packet memory, and ii) operate according to the configuration information.

Embodiment 3: The network device of embodiment 1, wherein the plurality of non-port components comprises: a first packet descriptor first-in-first-out (FIFO) queue; and a second packet descriptor FIFO queue; wherein the first packet descriptor FIFO queue is configured to send packet descriptors to the second packet descriptor FIFO queue by accessing the packet memory via one of the crossbars to store the packet descriptors in the packet memory for retrieval by the second packet descriptor FIFO queue, each packet descriptor comprising information corresponding to a packet, the information corresponding to the packet including information that was not within any headers of the packet when the packet was received by the network device and that is not within any headers of the packet when the packet is transmitted by the network device via one or more communication links among the plurality of communication links.

Embodiment 4: The network device of embodiment 1, wherein the plurality of non-port components comprises: a first non-port component configured to send management information to one or more of i) at least one port and ii) at least one other non-port component by accessing the packet memory via one of the crossbars to store the management information in the packet memory for retrieval by the one or more of i) the at least one port and ii) the at least one other non-port component, the management information comprising information corresponding to management of the network device; wherein the one or more of i) the at least one port and ii) the at least one other non-port component are

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configured to access the packet memory via at least one of the crossbars to retrieve the management information from the packet memory.

Embodiment 5: The network device of embodiment 1, wherein the plurality of non-port components comprises: a first non-port component configured to send debug information to one or more of i) at least one port and ii) at least one other non-port component by accessing the packet memory via one of the crossbars to store the debug information in the packet memory for retrieval by the one or more of i) the at least one port and ii) the at least one other non-port component, the debug information comprising information corresponding to debugging operation of the network device; wherein the one or more of i) the at least one port and ii) the at least one other non-port component are configured to access the packet memory via at least one of the crossbars to retrieve the debug information from the packet memory.

Embodiment 6: The network device of embodiment 1, further comprising: a memory management controller configured to allocate memory resources of the packet memory for storing packet data and to allocate memory resources of the packet memory for storing the non-packet data.

Embodiment 7: The network device of embodiment 6, wherein the memory management controller is further configured to: for respective pairs of non-port components of the network device, manage respective linked-lists of buffers in the packet memory that are configured to store respective non-packet data being communicated between the respective pairs of non-port components of the network device.

Embodiment 8: The network device of embodiment 1, further comprising: logic circuitry configured to determine when the packet memory is capable of providing reliable communication of non-packet data and to generate a signal that indicates when the packet memory is capable of providing reliable communication of non-packet data; wherein a first non-port component of the network device is configured to: when the signal indicates that the packet memory is capable of providing reliable communication of non-packet data, store non-packet data to the packet memory, and when the signal indicates that the packet memory is not capable of providing reliable communication of non-packet data, refrain from storing the non-packet data to the packet memory.

Embodiment 9: The network device of embodiment 1, wherein: a first non-port component of the network device includes first logic circuitry that is configured to aggregate multiple units of non-packet information into an aggregated set of non-packet information, and store the aggregated set of non-packet information in the packet memory; and a second non-port component of the network device includes second logic circuitry that is configured to unpack multiple units of non-packet information from the aggregated set of non-packet information that was retrieved from the packet memory.

Embodiment 10: The network device of embodiment 1, wherein the non-packet information comprises one or more of: management information that is used by a component of the network device to manage operation of the network device, control information that is configured to control operation of the component, configuration information that is used to configure the component of the network device, an indication of a location in the packet memory at which a packet is stored, an identifier of a port at which the packet was received, an indicator of a time at which the packet was received by the network device, an indicator of one or more ports via which the packet is to be transmitted, an indicator of whether the packet is to be dropped, an indicator of

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whether a tunneling header is to be added to the packet, an indicator of whether a tunneling header is to be removed from the packet, an indicator of whether a timestamp is to be added to the packet prior to transmitting the packet.

Embodiment 11: A method for communicating information within a network device, the method comprising: forwarding packets among a plurality of ports of the network device based on forwarding decisions made by a packet processor of the network device, the packets having been received via communication links coupled to the plurality of ports and subsequently transmitted by the network device via the plurality of ports, wherein forwarding the packets among the plurality of ports comprises storing at least payloads of the packets in a packet memory coupled to the plurality of ports until the packets are transmitted by the network device via ports of the network device, including i) accessing the packet memory via a plurality of crossbars coupled to the packet memory to write the packets to the packet memory, and ii) accessing the packet memory via the plurality of crossbars to read the packets from the packet memory; and communicating non-packet data within the network device, including i) accessing the packet memory via the plurality of crossbars to write the non-packet data to the packet memory, and ii) accessing the packet memory via the plurality of crossbars to read the non-packet data from the packet memory, the non-packet data comprising information not included in any packets i) received or transmitted by the network device and ii) for which the packet processor makes a forwarding decision.

Embodiment 12: The method of embodiment 11, wherein communicating non-packet data within the network device comprises: accessing, by a processor of the network device, the packet memory via the plurality of crossbars to write configuration information to the packet memory, the configuration information including information that i) controls operation of one or more of a) at least one port and b) at least one other non-port component of the network device, and ii) is not transmitted by any communication links communicatively coupled to any ports of the network device; and accessing, by the one or more of i) the at least one port and ii) the at least one other non-port component of the network device, the packet memory via the plurality of crossbars to retrieve the configuration information from the packet memory; wherein the method further comprises, after retrieving the configuration information from the packet memory, the one or more of i) the at least one port and ii) the at least one other non-port component operating according to the configuration information.

Embodiment 13: The method of embodiment 11, wherein communicating non-packet data within the network device comprises: accessing, by a first packet descriptor first-in-first-out (FIFO) queue of the network device, the packet memory via the plurality of crossbars to write packet descriptors to the packet memory, each packet descriptor comprising information corresponding to a packet, the information corresponding to the packet including information that was not within any headers of the packet when the packet was received by the network device and that is not within any headers of the packet when the packet is transmitted by the network device via one or more communication links among the plurality of communication links; and accessing, by a second packet descriptor FIFO queue of the network device, the packet memory via the plurality of crossbars to retrieve the packet descriptors from the packet memory.

Embodiment 14: The method of embodiment 11, wherein communicating non-packet data within the network device

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comprises: accessing, by a first non-port component of the network device, the packet memory via the plurality of crossbars to write management information to the packet memory, the management information comprising information corresponding to management of the network device; and accessing, by one or more of i) at least one port and ii) at least one other non-port component of the network device, the packet memory via the plurality of crossbars to retrieve the management information from the packet memory.

Embodiment 15: The method of embodiment 11, wherein communicating non-packet data within the network device comprises: accessing, by a first non-port component of the network device, the packet memory via the plurality of crossbars to write debug information to the packet memory, the debug information comprising information corresponding to debugging operation of the network device; and accessing, by one or more of i) at least one port and ii) at least one other non-port component of the network device, the packet memory via the plurality of crossbars to retrieve the debug information from the packet memory.

Embodiment 16: The method of embodiment 11, further comprising: allocating, by a memory management controller, memory resources of the packet memory for storing packet data; and allocating, by the memory management controller, memory resources of the packet memory for storing the non-packet data.

Embodiment 17: The method of embodiment 16, further comprising: for respective pairs of non-port components of the network device, managing, by the memory management controller, respective linked-lists of buffers in the packet memory that are configured to store respective non-packet data being communicated between the respective pairs of non-port components of the network device.

Embodiment 18: The method of embodiment 11, further comprising: generating, by logic circuitry, a signal that indicates when the packet memory is capable of providing reliable communication of non-packet data; when the signal indicates that the packet memory is capable of providing reliable communication of non-packet data, storing, by a first non-port component of the network device, non-packet data to the packet memory; and when the signal indicates that the packet memory is not capable of providing reliable communication of non-packet data, refraining, by the first non-port component, from storing the non-packet data to the packet memory.

Embodiment 19: The method of embodiment 11, further comprising: aggregating, by first logic circuitry of a first non-port component of the network device, multiple units of non-packet information into an aggregated set of non-packet information; storing the aggregated set of non-packet information in the packet memory; and unpacking, by second logic circuitry of a second non-port component of the network device, multiple units of non-packet information from the aggregated set of non-packet information that was retrieved from the packet memory.

Embodiment 20: The embodiment of embodiment 11, wherein the non-packet information comprises one or more of: management information that is used by a component of the network device to manage operation of the network device, control information that is configured to control operation of the component, configuration information that is used to configure the component of the network device, an indication of a location in the packet memory at which a packet is stored, an identifier of a port at which the packet was received, an indicator of a time at which the packet was received by the network device, an indicator of one or more ports via which the packet is to be transmitted, an indicator

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of whether the packet is to be dropped, an indicator of whether a tunneling header is to be added to the packet, an indicator of whether a tunneling header is to be removed from the packet, an indicator of whether a timestamp is to be added to the packet prior to transmitting the packet.

At least some of the various blocks, operations, and techniques described above may be implemented utilizing hardware, a processor executing firmware instructions, a processor executing software instructions, or any combination thereof. When implemented utilizing a processor executing software or firmware instructions, the software or firmware instructions may be stored in any suitable computer readable memory such as a random-access memory (RAM), a read only memory (ROM), a flash memory, etc. The software or firmware instructions may include machine readable instructions that, when executed by one or more processors, cause the one or more processors to perform various acts.

When implemented in hardware, the hardware may comprise one or more of discrete components, an integrated circuit, an application-specific integrated circuit (ASIC), a programmable logic device (PLD), etc.

While the present invention has been described with reference to specific examples, which are intended to be illustrative only and not to be limiting of the invention, changes, additions and/or deletions may be made to the disclosed embodiments without departing from the scope of the invention.

What is claimed is:

1. A network device, comprising:

- a plurality of ports configured to be communicatively coupled to a plurality of communication links;
- a packet processor configured to process at least headers of packets received via the ports of the network device to make forwarding decisions regarding the packets;
- a packet memory coupled to the plurality of ports, the packet memory configured to store at least payloads of the packets received via the ports of the network device until the packets are transmitted by the network device via the ports of the network device;
- a plurality of crossbars coupled to the packet memory, each crossbar configured to provide a respective set of ports of the network device with read access and write access to the packet memory; and
- a plurality of non-port components associated with processing the packets received by the network device, the plurality of non-port components coupled to the plurality of crossbars, each one or more of the non-port components being configured to
 - determine when the packet memory is capable of providing reliable communication of the non-packet data, and
 - in response to determining that the packet memory is capable of providing the reliable communication of the non-packet data, send the non-packet data to one or more of i) at least one port and ii) at least one other non-port component, by storing the non-packet data in the packet memory for retrieval by the one or more of i) the at least one port and ii) the at least one other non-port component, the non-packet data comprising information not included in any packet i) received or transmitted by the network device and ii) for which the packet processor makes the forwarding decisions.

2. The network device of claim 1, wherein the plurality of non-port components comprises:

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a processor configured to send configuration information to the one or more of i) the at least one port and ii) the at least one other non-port component by accessing the packet memory via one of the crossbars to store the configuration information in the packet memory for retrieval by the one or more of i) the at least one port and ii) the at least one other non-port component, the configuration information including information that i) controls operation of the one or more of a) the at least one port and b) the at least one other non-port component and ii) is not transmitted by any communication links communicatively coupled to any ports of the network device; and

wherein the one or more of i) the at least one port and ii) the at least one other non-port component are configured to: i) access the packet memory via at least one of the crossbars to retrieve the configuration information from the packet memory, and ii) operate according to the configuration information.

3. The network device of claim 1, wherein the plurality of non-port components comprises:

- a first packet descriptor first-in-first-out (FIFO) queue; and
- a second packet descriptor FIFO queue;

wherein the first packet descriptor FIFO queue is configured to send packet descriptors to the second packet descriptor FIFO queue by accessing the packet memory via one of the crossbars to store the packet descriptors in the packet memory for retrieval by the second packet descriptor FIFO queue, each packet descriptor comprising information corresponding to a packet of the packets received via the ports, the information in each packet descriptor corresponding to the packet including information that was not within any headers of the packet when the packet was received by the network device and that is not within any headers of the packet when the packet is transmitted by the network device via one or more communication links among the plurality of communication links.

4. The network device of claim 1, wherein the plurality of non-port components comprises:

- a first non-port component configured to send management information to the one or more of i) the at least one port and ii) the at least one other non-port component by accessing the packet memory via one of the crossbars to store the management information in the packet memory for retrieval by the one or more of i) the at least one port and ii) the at least one other non-port component, the management information comprising information corresponding to management of the network device; and

wherein the one or more of i) the at least one port and ii) the at least one other non-port component are configured to access the packet memory via at least one of the crossbars to retrieve the management information from the packet memory.

5. The network device of claim 1, wherein the plurality of non-port components comprises:

- a first non-port component configured to send debug information to the one or more of i) the at least one port and ii) the at least one other non-port component by accessing the packet memory via one of the crossbars to store the debug information in the packet memory for retrieval by the one or more of i) the at least one port and ii) the at least one other non-port component, the debug information comprising information corresponding to debugging operation of the network device; and

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wherein the one or more of i) the at least one port and ii) the at least one other non-port component are configured to access the packet memory via at least one of the crossbars to retrieve the debug information from the packet memory.

6. The network device of claim 1, further comprising:

- a memory management controller configured to allocate memory resources of the packet memory for storing packet data and to allocate memory resources of the packet memory for storing the non-packet data.

7. The network device of claim 6, wherein the memory management controller is further configured to:

- for respective pairs of non-port components of the network device, manage respective linked-lists of buffers in the packet memory that are configured to store respective non-packet data being communicated between the respective pairs of non-port components of the network device.

8. The network device of claim 1, further comprising:

- logic circuitry configured to determine when the packet is capable of providing the reliable communication of the non-packet data and to generate a signal that indicates when the packet memory is capable of providing the reliable communication of the non-packet data; and

wherein a first non-port component of the network device is configured to:

- when the signal indicates that the packet memory is capable of providing the reliable communication of the non-packet data, store the non-packet data to the packet memory, and when the signal indicates that the packet memory is not capable of providing the reliable communication of the non-packet data, refrain from storing the non-packet data to the packet memory.

9. The network device of claim 1, wherein:

- a first non-port component of the network device includes first logic circuitry that is configured to aggregate multiple units of non-packet information into an aggregated set of non-packet information, and store the aggregated set of non-packet information in the packet memory; and
- a second non-port component of the network device includes second logic circuitry that is configured to unpack the multiple units of non-packet information from the aggregated set of non-packet information that was retrieved from the packet memory.

10. The network device of claim 1, wherein the non-packet information comprises one or more of:

- management information that is used by a component of the network device to manage operation of the network device, control information that is configured to control operation of the component, configuration information that is used to configure the component of the network device, an indication of a location in the packet memory at which a packet is stored, an identifier of a port at which the packet was received, an indicator of a time at which the packet was received by the network device, an indicator of one or more ports via which the packet is to be transmitted, an indicator of whether the packet is to be dropped, an indicator of whether a tunneling header is to be added to the packet, an indicator of whether a tunneling header is to be removed from the packet, and an indicator of whether a timestamp is to be added to the packet prior to transmitting the packet.

11. A method for communicating information within a network device, the method comprising:

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forwarding packets among a plurality of ports of the network device based on forwarding decisions made by a packet processor of the network device, the packets having been received via communication links coupled to the plurality of ports and subsequently transmitted by the network device via the plurality of ports, wherein forwarding the packets among the plurality of ports comprises storing at least payloads of the packets in a packet memory coupled to the plurality of ports until the packets are transmitted by the network device via the ports of the network device, including i) accessing the packet memory via a plurality of crossbars coupled to the packet memory to write the packets to the packet memory, and ii) accessing the packet memory via the plurality of crossbars to read the packets from the packet memory; and

communicating non-packet data within the network device, including

i) determining when the packet memory is capable of providing reliable communication of the non-packet data,

in response to determining that the packet memory is capable of providing the reliable communication of the non-packet data, accessing the packet memory via the plurality of crossbars to write the non-packet data to the packet memory, and

iii) accessing the packet memory via the plurality of crossbars to read the non-packet data from the packet memory, the non-packet data comprising information not included in any packets i) received or transmitted by the network device and ii) for which the packet processor makes the forwarding decisions.

12. The method of claim 11, wherein communicating non-packet data within the network device comprises:

accessing, by a processor of the network device, the packet memory via the plurality of crossbars to write configuration information to the packet memory, the configuration information including information that i) controls operation of one or more of a) at least one port and b) at least one other non-port component of the network device, and ii) is not transmitted by any communication links communicatively coupled to any ports of the network device; and

accessing, by the one or more of i) the at least one port and ii) the at least one other non-port component of the network device, the packet memory via the plurality of crossbars to retrieve the configuration information from the packet memory; and

wherein the method further comprises, after retrieving the configuration information from the packet memory, the one or more of i) the at least one port and ii) the at least one other non-port component operating according to the configuration information.

13. The method of claim 11, wherein communicating non-packet data within the network device comprises:

accessing, by a first packet descriptor first-in-first-out (FIFO) queue of the network device, the packet memory via the plurality of crossbars to write packet descriptors to the packet memory, each packet descriptor comprising information corresponding to a packet of the packets received via the ports, the information in each packet descriptor corresponding to the packet including information that was not within any headers of the packet when the packet was received by the network device and that is not within any headers of the packet when the packet is transmitted by the network

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device via one or more communication links among the plurality of communication links; and

accessing, by a second packet descriptor FIFO queue of the network device, the packet memory via the plurality of crossbars to retrieve the packet descriptors from the packet memory.

14. The method of claim 11, wherein communicating non-packet data within the network device comprises:

accessing, by a first non-port component of the network device, the packet memory via the plurality of crossbars to write management information to the packet memory, the management information comprising information corresponding to management of the network device; and

accessing, by one or more of i) the at least one port and ii) the at least one other non-port component of the network device, the packet memory via the plurality of crossbars to retrieve the management information from the packet memory.

15. The method of claim 11, wherein communicating non-packet data within the network device comprises:

accessing, by a first non-port component of the network device, the packet memory via the plurality of crossbars to write debug information to the packet memory, the debug information comprising information corresponding to debugging operation of the network device; and

accessing, by one or more of i) the at least one port and ii) the at least one other non-port component of the network device, the packet memory via the plurality of crossbars to retrieve the debug information from the packet memory.

16. The method of claim 11, further comprising:

allocating, by a memory management controller, memory resources of the packet memory for storing packet data; and

allocating, by the memory management controller, memory resources of the packet memory for storing the non-packet data.

17. The method of claim 16, further comprising:

for respective pairs of non-port components of the network device, managing, by the memory management controller, respective linked-lists of buffers in the packet memory that are configured to store respective non-packet data being communicated between the respective pairs of non-port components of the network device.

18. The method of claim 11, further comprising:

generating, by logic circuitry, a signal that indicates when the packet memory is capable of providing reliable communication of the non-packet data;

when the signal indicates that the packet memory is capable of providing the reliable communication of the non-packet data, storing, by a first non-port component of the network device, the non-packet data to the packet memory; and

when the signal indicates that the packet memory is not capable of providing the reliable communication of the non-packet data, refraining, by the first non-port component, from storing the non-packet data to the packet memory.

19. The method of claim 11, further comprising:

aggregating, by first logic circuitry of a first non-port component of the network device, multiple units of non-packet information into an aggregated set of non-packet information;

storing the aggregated set of non-packet information in the packet memory; and unpacking, by second logic circuitry of a second non-port component of the network device, the multiple units of non-packet information from the aggregated set of non-packet information that was retrieved from the packet memory. 5

20. The method of claim 11, wherein the non-packet information comprises one or more of:

management information that is used by a component of the network device to manage operation of the network device, control information that is configured to control operation of the component, configuration information that is used to configure the component of the network device, an indication of a location in the packet memory at which a packet is stored, an identifier of a port at which the packet was received, an indicator of a time at which the packet was received by the network device, an indicator of one or more ports via which the packet is to be transmitted, an indicator of whether the packet is to be dropped, an indicator of whether a tunneling header is to be added to the packet, an indicator of whether a tunneling header is to be removed from the packet, and an indicator of whether a timestamp is to be added to the packet prior to transmitting the packet. 25

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